

Henry W. Leung, Ph.D.

Machine Learning & Data Scientist

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SUMMARY

Machine Learning and Data Scientist with 6+ years of experience adapting ML techniques to large-scale datasets and 9+ years of software development in Python/C. Expertise in delivering data-driven solutions and insights for real-world applications.

PROFESSIONAL EXPERIENCE

Data Sciences Institute, University of Toronto

Sept. 2023 – Oct. 2024

Data Sciences Doctoral Fellow

- Developed applications of **Transformers** architecture with a **denoising diffusion** probabilistic head as density function emulator for tabular data to improve non-Gaussian uncertainty estimation. Applicable to uncertainty quantification in finance and healthcare.
- Curated multiple datasets for a 100 TB machine learning-ready dataset hosted on **huggingface**, for the next-generation large ML models in physical science and industry.

University of Toronto

Sept. 2019 – Oct. 2024

Graduate Researcher

- Developed a **self-supervised** foundation model in **PyTorch** with a novel token-scalar embedding technique, outperforming XGBoost by 6% in multiple predictive tasks. Presented at NeurIPS and ICML for advancing structured tabular data learning.
- Designed an **unsupervised** encoder-decoder in **TensorFlow** to address a data-rich label-scarce challenge, by reducing data dimensionality by 1,000x through physics-guided techniques and label regression in the low-dimensional latent space.
- Public product release of ML-derived parameters estimation models with uncertainty quantification, achieving over a 10% accuracy improvement for **low signal-to-noise** data and a 100x speedup compared to non-ML pipelines.
- Built a Flask web app integrating an LLM for seamless natural language interaction with scientific models, demonstrating LLM-driven interaction with specialized tools in workflows.
- Implemented supervised **CNNs with dropout variational inference** for spectral data analysis, contributing to one of the **first direct mappings** of the Milky Way's inner structure.
- Developed and maintained open-source software that are **well-documented** with rich set of examples and **thoroughly tested** with more than 80% code coverages, as well as contributed to other open-source projects.

TECHNICAL CONTRIBUTIONS

Below is a summary of key research contributions, highlighting those in machine learning conferences:

NeurIPS (Oral in 2023, Collaboration Poster in 2024), ICML (Poster in 2024), AI for Astronomy (Oral in 2019)

- Research in **ML adaptation** and **foundation models** for scientific and structured datasets with Transformers and diffusion models. ML-driven predictive modeling robust to noise to extract insights from large-scale datasets with applications in finance, healthcare, and physical sciences
- Discovery enabled by **data-driven insights**, including anomaly detection and ML-derived parameter estimation, enabling analyses otherwise difficult with traditional methods.

First/Second author on 9 refereed papers with 770+ citations among 16 refereed papers with 3880+ citations (h-index=12).

EDUCATION

Ph.D., Astronomy & Astrophysics, University of Toronto

Sept. 2020 – Oct. 2024

M.Sc., Astronomy & Astrophysics, University of Toronto

Sept. 2019 – Aug. 2020

H.B.Sc., Physics & Astronomy, University of Toronto

Sept. 2014 – Aug. 2019

SKILLS & SOFTWARE

Programming Languages: Python, C (proficient), C++, SQL (intermediate), Rust (beginner).**Frameworks/Packages:** Proficient in PyTorch, TensorFlow, Scikit-Learn, NumPy, SciPy, Pandas, PySpark, Jupyter Notebook, Matplotlib, SQLite, PostgreSQL, Docker, SSH, Git, Bash, Slurm, Hugo, SCSS, Node.js. Some experiences in ArcGIS and SolidWorks.

My open-source software packages used by the community that are well tested with continuous integration and well documented with docstrings and user guides, includes:

- **astroNN** (195 stars) – **Deep learning framework** with Keras for astronomical data, supporting TensorFlow and PyTorch.
- **MyGaiaDB** – **SQLite-based database** package for managing and querying of astronomical data locally.
- **Galaxy10** – **Benchmark dataset** for CNN-based galaxy classification, used in ML research and education.